16

Non-neutral plasmas

16.1 Introduction

Conventional plasmas, the main topic of this book, consist of a quasi-neutral collection of mutually interacting ions and electrons. This description applies to the vast majority of entities called plasma (e.g., fusion, industrial, propulsion, ionospheric, magnetospheric, interplanetary, solar, and astrophysical plasmas). Non-neutral plasmas are an exception to this taxonomy and, not surprisingly, have certain behaviors that differ from conventional plasmas (Malmberg 1992, Davidson 2001). Applications of non-neutral plasmas include microwave oscillators and amplifiers, simulating the vortex dynamics of ideal two-dimensional hydrodynamics, testing basic nonlinear concepts, and confining anti-particles.

16.2 Brillouin flow

Using a cylindrical coordinate system $\{r, \theta, z\}$ we first consider the forces acting on an infinitely long, azimuthally symmetric cylindrical cloud of cold charged particles all having the *same* polarity. Because of the mutual electrostatic repulsion of the same-sign charged particles, the cloud of particles will expand continuously in the r direction and so will not be in radial equilibrium. The cloud would be even further from equilibrium if, in addition, it were to rotate with azimuthal velocity $\mathbf{u} = u_\theta\hat{\theta}$, because rotation would provide a centrifugal force mu_θ^2/r, which would add to the radially outwards electrostatic force and so increase the rate of radially outward expansion.

However, if this rotating cloud of same-sign charged particles were immersed in a uniform axial magnetic field $\mathbf{B} = B\hat{z}$, then the magnetic force $\sim q\mathbf{u} \times \mathbf{B}$ would push the charged particles radially inwards, counteracting both the electrostatic repulsion and the centrifugal force. This situation is shown in Fig. 16.1 for the situation where the cloud consists of electrons (Malmberg and de Grassie 1975). Negatively biased electrodes are used at the ends to prevent cloud expansion in

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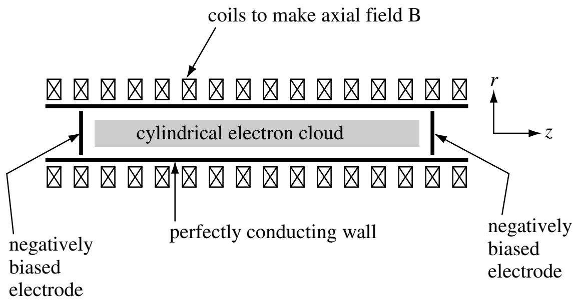

Fig. 16.1 Pure electron plasma configuration. Magnetic field $\mathbf{B} = B\hat{z}$ produced by coils, negatively biased electrodes on ends prevent axial expansion. Plasma radius is r_p, wall radius is a.

the z direction; if the cloud were made of ions, then the end electrodes would be biased positively.

Because there is only one charge species, there is no frictional drag due to collisions with a species of opposite polarity and, because the plasma is cold, the pressure is zero. The radial component of the fluid equation of motion Eq. (2.27) thus reduces to a simple competition between the electrostatic, magnetic, and centrifugal forces, namely

$$0 = q\left(E_r + u_\theta B_z\right) + \frac{m u_\theta^2}{r}. \tag{16.1}$$

Because of the assumed cylindrical and azimuthal symmetry, Poisson's equation reduces to

$$\frac{1}{r}\frac{\partial}{\partial r}\left(rE_r\right) = \frac{n(r)q}{\varepsilon_0}, \tag{16.2}$$

which can be integrated to give

$$E_r = \frac{q}{\varepsilon_0}\frac{1}{r}\int_0^r n(r')r'\mathrm{d}r'. \tag{16.3}$$

We will assume that the radius of the non-neutral plasma is r_p and that the non-neutral plasma is surrounded by a perfectly conducting wall of radius a, where a is larger than r_p so there is a vacuum region between the plasma and the wall. In the special case of uniform density up to the plasma radius r_p, Eq. (16.3) may be evaluated to give

$$E_r = \begin{cases} \dfrac{nq}{2\varepsilon_0} r \text{ for } r \leq r_p, \text{ plasma region} \\ \dfrac{nq}{2\varepsilon_0}\dfrac{r_p^2}{r} \text{ for } r_p \leq r \leq a, \text{ vacuum region.} \end{cases} \tag{16.4}$$

Thus, in the plasma Eq. (16.1) becomes

$$u_\theta^2 + u_\theta r\omega_c + \omega_p^2 r^2/2 = 0, \tag{16.5}$$

a quadratic equation for u_θ. It is convenient to express the two roots of this equation in terms of angular velocities $\omega_0 = u_\theta/r$ so

$$\omega_0 = \frac{-\omega_c \pm \sqrt{\omega_c^2 - 2\omega_p^2}}{2}. \tag{16.6}$$

Since ω_0 is independent of r, the non-neutral plasma rotates as a rigid body. This is a special case resulting from the assumption of a uniform density profile. In the more general case of a non-uniform density profile (to be discussed later), the rotation velocity is sheared so ω_0 varies with r. The two roots in Eq. (16.6) coalesce at $\omega_p^2 = \omega_c^2/2$. This point of coalescence is called the Brillouin limit and it is seen that real roots ω_0 exist only if the density is sufficiently low for ω_p^2 to be below this limit.

We will consider the situation typical for non-neutral plasma experiments where the density is well below the Brillouin limit so the two roots are well separated and given by

$$\omega_{0-} \simeq -\omega_c\left(1 + \omega_p^2/2\omega_c^2\right) \tag{16.7}$$

$$\omega_{0+} \simeq \omega_p^2/2\omega_c; \tag{16.8}$$

the plus and minus signs refer to the choice of signs in Eq. (16.6). The large root ω_{0-} is near the cyclotron frequency and the small root ω_{0+} is much smaller than the plasma frequency since $\omega_p \ll \omega_c$. The small root is called the diocotron frequency.

The rotation of the non-neutral plasma in the magnetic field provides an inward force balancing the radially outward electrostatic force. There also exist axial electrostatic forces due to the mutual electrostatic repulsion between the same-sign charges and these forces would cause the plasma to expand axially. Since these axial forces cannot be balanced magnetically, the axial forces are balanced by biased electrodes at the ends of the plasma. The electrodes have the same polarity as the plasma, thereby providing a potential well in the axial direction. Any particle attempting to escape axially is repelled by forces due to the repulsive bias on an end electrode and is reflected back into the main plasma cloud before it can reach the end electrode.

The fast motion of the charged particles in the axial direction smears out axial structure so, to first approximation, the non-neutral plasma can be considered axially uniform. The system then depends only on r and θ and a pair of deceptively simple-looking coupled equations relates the electrostatic potential ϕ and the density n. These two coupled equations govern not only the equilibrium but also the surprisingly rich low-frequency dynamics of a non-neutral plasma. Because the electric field is electrostatic and the magnetic field is a uniform vacuum field, the $\mathbf{E} \times \mathbf{B}$ drift

$$\mathbf{u} = \frac{-\nabla\phi \times \mathbf{B}}{B^2} \tag{16.9}$$

is incompressible, i.e., $\nabla \cdot \mathbf{u} = 0$. Because of this, the continuity equation, Eq. (2.19), reduces to a simple convection of density with flow, namely

$$\frac{\partial n}{\partial t} + \mathbf{u} \cdot \nabla n = 0. \tag{16.10}$$

Using Eq. (16.9) to substitute for $\mathbf{u}$, this can be written as

$$\frac{\partial n}{\partial t} = \frac{\nabla\phi \times \mathbf{B}}{B^2} \cdot \nabla n. \tag{16.11}$$

Equation (16.11) and Poisson's equation

$$\nabla^2\phi = -\frac{nq}{\varepsilon_0} \tag{16.12}$$

provide two coupled equations in n and ϕ and are the governing equations for the configuration. Although Eqs. (16.11) and (16.12) appear simple, they actually allow quite complex behavior, which will be discussed in the remainder of this chapter. An interesting feature of these equations is that they have no mass dependence, a consequence of ignoring the centrifugal force term, which affects only the high-frequency branch, Eq. (16.7). Thus, the low-frequency dynamics described by Eqs. (16.11) and (16.12) can be thought of as the dynamics of a massless, incompressible fluid governed by a combination of $\mathbf{E} \times \mathbf{B}$ drifts and Poisson's equation.

16.3 Isomorphism to incompressible 2-D hydrodynamics

Let us put plasma physics aside for a moment and consider the equations governing an incompressible two-dimensional fluid. Incompressibility means that the mass density ρ is constant and uniform so the continuity equation reduces to

$$\nabla \cdot \mathbf{u} = 0. \tag{16.13}$$

Using the vector identity $\mathbf{u} \cdot \nabla\mathbf{u} = \nabla u^2/2 - \mathbf{u} \times \nabla \times \mathbf{u}$, the fluid equation of motion can be written as

$$\rho\left(\frac{\partial \mathbf{u}}{\partial t} + \frac{1}{2}\nabla u^2 - \mathbf{u} \times \nabla \times \mathbf{u}\right) = -\nabla P. \tag{16.14}$$

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Taking the curl of this equation and defining the vorticity vector $\mathbf{\Omega} = \nabla \times \mathbf{u}$ gives

$$\frac{\partial \mathbf{\Omega}}{\partial t} = \nabla \times (\mathbf{u} \times \mathbf{\Omega}), \tag{16.15}$$

which has the same form as the ideal MHD induction equation, Eq. (2.82), and so can be interpreted as indicating that the vorticity is frozen into the convecting fluid (Kelvin vorticity theorem). Using cylindrical coordinates to express the velocity as $\mathbf{u} = u_r(r, \theta)\hat{r} + u_\theta(r, \theta)\hat{\theta}$ it is seen that

$$\nabla \times \mathbf{u} = \hat{z}\left(\frac{1}{r}\frac{\partial}{\partial r}(ru_\theta) - \frac{1}{r}\frac{\partial u_r}{\partial \theta}\right). \tag{16.16}$$

The vorticity vector therefore lies in the z direction and may be written as

$$\mathbf{\Omega} = \Omega\hat{z}. \tag{16.17}$$

The z component of Eq. (16.15) is

$$\begin{aligned}\frac{\partial \Omega}{\partial t} &= \hat{z} \cdot \nabla \times (\mathbf{u} \times \Omega\hat{z}) \\ &= \nabla \cdot ((\mathbf{u} \times \Omega\hat{z}) \times \hat{z}) \\ &= -\nabla \cdot (\mathbf{u}\Omega) \\ &= -\mathbf{u} \cdot \nabla\Omega. \end{aligned} \tag{16.18}$$

The incompressibility condition prescribed by Eq. (16.13) allows the velocity to be written in terms of a stream-function ψ,

$$\mathbf{u} = -\nabla\psi \times \hat{z}, \tag{16.19}$$

in which case Eq. (16.18) becomes

$$\frac{\partial \Omega}{\partial t} = \nabla\psi \times \hat{z} \cdot \nabla\Omega. \tag{16.20}$$

However, the z component of the curl of Eq. (16.19) can also be expressed in terms of ψ since

$$\begin{aligned}\Omega &= \hat{z} \cdot \nabla \times \mathbf{u} \\ &= -\hat{z} \cdot \nabla \times (\nabla\psi \times \hat{z}) \\ &= -\nabla \cdot ((\nabla\psi \times \hat{z}) \times \hat{z}) \\ &= \nabla^2\psi. \end{aligned} \tag{16.21}$$

Table 16.1

Non-neutral plasma	Incompressible 2-D fluid
$\dfrac{\partial n}{\partial t} = \dfrac{\nabla\phi \times \hat{z}}{B} \cdot \nabla n$	$\dfrac{\partial \Omega}{\partial t} = \nabla\psi \times \hat{z} \cdot \nabla\Omega$
$\nabla^2\phi = -\dfrac{nq}{\varepsilon_0}$	$\nabla^2\psi = \Omega$

There is thus an exact isomorphism between the non-neutral plasma equations and the equations describing a 2-D incompressible fluid; this isomorphism is given in Table 16.1

These sets of equations can be made identical by setting

$$\psi = \frac{\phi}{B}, \;\; \Omega = -\frac{nq}{\varepsilon_0 B}. \tag{16.22}$$

Thus, density corresponds to vorticity and electrostatic potential corresponds to stream-function.

A non-neutral plasma can therefore be used as an analog computer for investigating the behavior of an inviscid incompressible 2-D fluid (Driscoll and Fine 1990). This is quite useful because it is difficult to make a real fluid act in a truly two-dimensional inviscid fashion whereas it is relatively easy to make a non-neutral plasma do so. While real fluids are three dimensional, it is nevertheless very useful to develop an understanding for two-dimensional dynamics since this understanding can be of considerable help for understanding three-dimensional dynamics. The plasma analog contains all the nonlinear vortex interactions intrinsic to the 2-D fluid problem. Besides serving as an analog computer for investigations of 2-D fluid dynamics, non-neutral plasmas have also been successfully used as a method for trapping antimatter (Surko, Leventhal, and Passner 1989).

16.4 Near-perfect confinement

A curious feature of non-neutral plasmas is that collisions do not degrade confinement so long as axisymmetry is maintained (O'Neil 1995). Thus, in a pure electron plasma, electron–electron collisions do not cause leakage of the plasma out of the trap (loss of confinement comes only from collisions with neutrals and this can be minimized by using good vacuum techniques). To see this, consider the canonical angular momentum of the ith particle

$$P_{\theta i} = m r_i v_{\theta i} + q r_i A_{\theta i}. \tag{16.23}$$

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A collision between two identical charged particles will conserve the total angular momentum of the two particles and so the total angular momentum of all the particles is

$$\sum_{i=1}^{N} P_{\theta i} = \sum_{i=1}^{N} \left(m r_i v_{\theta_i} + q r_i A_{\theta i} \right) = const. \tag{16.24}$$

even when there are electron–electron collisions. The vector potential for the magnetic field $\mathbf{B} = B\hat{z}$ is $\mathbf{A} = \hat{\theta} Br/2$, and if the magnetic field is sufficiently strong, the inertial term in Eq. (16.24) can be dropped, so that conservation of total canonical angular momentum reduces to the simple relationship

$$P_\theta = \sum_{i=1}^{N} P_{\theta i} \simeq \frac{q_e B}{2} \sum_{i=1}^{N} r_i^2 = const. \tag{16.25}$$

Equation (16.25) constrains the plasma from moving radially outwards in an axisymmetric fashion. Thus, interparticle collisions cannot make the plasma diffuse to the wall and so a collisional plasma is perfectly confined so long as axisymmetry is maintained.

An alternative interpretation can be developed by considering how collisions cause axisymmetric outward diffusion in a conventional (i.e., electron–ion) plasma. This was discussed in Section 2.8 and will be briefly reviewed here emphasizing the difference between a conventional plasma and a non-neutral plasma. The effect of collisions in a conventional plasma can be seen by considering the steady-state azimuthal component of the resistive MHD Ohm's law. This azimuthal component is

$$-U_r B = \eta J_\theta = \left(\frac{m_e \nu_{ei}}{ne^2} \right) ne(u_{i\theta} - u_{e\theta}) = \frac{m_e \nu_{ei}}{e} (u_{i\theta} - u_{e\theta}) \tag{16.26}$$

and shows that axisymmetric radial flow to the wall results from collisions between *unlike* particles since

$$U_r = -\frac{m_e \nu_{ei}}{eB} (u_{i\theta} - u_{e\theta}). \tag{16.27}$$

Thus, radial transport requires momentum exchange between unlike species and this exchange occurs at a rate dictated by the collision frequency ν_{ei} and by the difference between electron and ion azimuthal velocities. If only one species exists, it is clearly impossible for momentum to be exchanged between unlike species and so there cannot be a net radial motion. A practical consequence of this result is that particle confinement in single species plasmas is orders of magnitude better than confinement in conventional quasi-neutral plasmas (hours/weeks compared to microseconds/seconds).

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16.5 Diocotron modes

Non-neutral plasmas support waves that differ significantly from the waves in a conventional, quasi-neutral plasma (Gould 1995). The theory of low-frequency linear waves in a cylindrical non-neutral plasma can be developed by linearizing the continuity equation Eq. (16.10) to obtain

$$\frac{\partial n_1}{\partial t} + \mathbf{u}_0 \cdot \nabla n_1 + \mathbf{u}_1 \cdot \nabla n_0 = 0. \tag{16.28}$$

Using Eq. (16.9) to give $\mathbf{u}_0$, $\mathbf{u}_1$, and using Poisson's equation, Eq. (16.12), to give n_0, n_1, results in the linear wave equation

$$\frac{\partial \nabla^2 \phi_1}{\partial t} - \frac{\nabla \phi_0 \times \mathbf{B}}{B^2} \cdot \nabla \nabla^2 \phi_1 - \frac{\nabla \phi_1 \times \mathbf{B}}{B^2} \cdot \nabla \nabla^2 \phi_0 = 0. \tag{16.29}$$

Because of the cylindrical geometry, it is convenient to decompose the potential perturbation ϕ_1 into azimuthal Fourier modes so

$$\phi_1(r, \theta, t) = \sum_{l=-\infty}^{\infty} \tilde{\phi}_l(r, t) \mathrm{e}^{\mathrm{i}l\theta}, \tag{16.30}$$

where

$$\tilde{\phi}_l(r, t) = \frac{1}{2\pi} \int_0^{2\pi} \mathrm{d}\theta \phi_1(r, \theta, t) \mathrm{e}^{-\mathrm{i}l\theta}. \tag{16.31}$$

The lth azimuthal mode is assumed to have a time dependence $\exp(-\mathrm{i}\omega_l t)$ so

$$\phi_1(r, \theta, t) = \sum_l \tilde{\phi}_l(r) \mathrm{e}^{\mathrm{i}l\theta - \mathrm{i}\omega_l t} \tag{16.32}$$

and the Fourier coefficient is given by

$$\tilde{\phi}_l(r) = \frac{1}{2\pi} \int_0^{2\pi} \mathrm{d}\theta \phi_1(r, \theta, t) \mathrm{e}^{-\mathrm{i}l\theta + \mathrm{i}\omega_l t}. \tag{16.33}$$

Using the equilibrium azimuthal velocity

$$u_{\theta 0}(r) = \frac{-\nabla \phi_0 \times \mathbf{B}}{B^2} \cdot \hat{\theta} \tag{16.34}$$

the temporal-azimuthal Fourier transform of Eq. (16.29) can be written as

$$\left(\omega_l - \frac{l u_{\theta 0}(r)}{r} \right) \nabla^2 \tilde{\phi}_l + \frac{l \tilde{\phi}_l}{rB} \frac{d}{dr} \nabla^2 \phi_0 = 0. \tag{16.35}$$

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Using the relations

$$\nabla^2\tilde{\phi}_l = \frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}\left(r\frac{\mathrm{d}\tilde{\phi}_l}{\mathrm{d}r}\right) - \frac{l^2}{r^2}\tilde{\phi}_l, \tag{16.36}$$

$$\nabla^2\phi_0 = \frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}\left(r\frac{\mathrm{d}\phi_0}{\mathrm{d}r}\right), \tag{16.37}$$

this temporal-azimuthal Fourier transform of Eq. (16.29) can be expanded as

$$\left(\omega_l - \frac{lu_{\theta 0}(r)}{r}\right)\left(\frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}\left(r\frac{\mathrm{d}\tilde{\phi}_l}{\mathrm{d}r}\right) - \frac{l^2}{r^2}\tilde{\phi}_l\right) + \frac{l\tilde{\phi}_l}{rB}\frac{\mathrm{d}}{\mathrm{d}r}\left(\frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}\left(r\frac{\mathrm{d}\phi_0}{\mathrm{d}r}\right)\right) = 0. \tag{16.38}$$

The equilibrium angular velocity is

$$\omega_0(r) = \frac{u_{\theta 0}(r)}{r} = \frac{1}{rB}\frac{\mathrm{d}\phi_0}{\mathrm{d}r} \tag{16.39}$$

and so Eq. (16.38) can be expressed as

$$(\omega - l\omega_0(r))\left(\frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}\left(r\frac{\mathrm{d}\tilde{\phi}_l}{\mathrm{d}r}\right) - \frac{l^2}{r^2}\tilde{\phi}_l\right) + \frac{l\tilde{\phi}_l}{r}\frac{\mathrm{d}}{\mathrm{d}r}\left(\frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2\omega_0(r)\right)\right) = 0, \tag{16.40}$$

where the l subscript has been dropped from ω.

Since Eq. (16.37) gives

$$\frac{\mathrm{d}\phi_0}{\mathrm{d}r} = -\frac{q}{\varepsilon_0 r}\int_0^r n_0(r')r'\mathrm{d}r', \tag{16.41}$$

the equilibrium angular velocity can be evaluated in terms of the equilibrium density to obtain

$$\omega_0(r) = -\frac{q}{\varepsilon_0 B r^2}\int_0^r n_0(r')r'\mathrm{d}r'. \tag{16.42}$$

Equation (16.42) reduces to the special case of rigid body rotation when the density is uniform, but in the more general case there is a shear in the angular velocity. Combining Eq. (16.39) and Poisson's equation shows that

$$\frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2\omega_0(r)\right) = \frac{1}{Br}\frac{\mathrm{d}}{\mathrm{d}r}\left(r\frac{\mathrm{d}\phi_0}{\mathrm{d}r}\right) = -\frac{nq}{\varepsilon_0 B} = -\frac{\omega_p^2(r)}{\omega_c}. \tag{16.43}$$

Furthermore, since

$$\frac{\mathrm{d}}{\mathrm{d}r}\left(\frac{1}{r}\frac{\mathrm{d}r^2}{\mathrm{d}r}\right) = 0, \tag{16.44}$$

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an arbitrary constant can be added to $\omega_0(r)$ in the last term in Eq. (16.40), which consequently can be written as

$$[\omega - l\omega_0(r)]\left(\frac{\mathrm{d}}{\mathrm{d}r}\left(r\frac{\mathrm{d}\tilde{\phi}_l}{\mathrm{d}r}\right) - \frac{l^2}{r}\tilde{\phi}_l\right) - \tilde{\phi}_l\frac{\mathrm{d}}{\mathrm{d}r}\left(\frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}\left[r^2\left(\omega - l\omega_0\right)\right]\right) = 0; \tag{16.45}$$

this is called the diocotron wave equation.

The diocotron wave equation can be written in a more symmetric form by defining

$$\begin{aligned} \Phi &= r\tilde{\phi}_l, \\ G &= r^2\left(\omega - l\omega_0(r)\right), \end{aligned} \tag{16.46}$$

and using

$$\frac{\mathrm{d}}{\mathrm{d}r}\left(r\frac{\mathrm{d}\tilde{\phi}_l}{\mathrm{d}r}\right) = r\frac{\mathrm{d}}{\mathrm{d}r}\left(\frac{1}{r}\frac{\mathrm{d}\Phi}{\mathrm{d}r}\right) + \frac{\Phi}{r^2} \tag{16.47}$$

to obtain

$$r\frac{\mathrm{d}}{\mathrm{d}r}\left(\frac{1}{r}\frac{\mathrm{d}\Phi}{\mathrm{d}r}\right) + \left(1 - l^2\right)\Phi - \Phi\frac{r}{G}\frac{\mathrm{d}}{\mathrm{d}r}\left(\frac{1}{r}\frac{\mathrm{d}G}{\mathrm{d}r}\right) = 0. \tag{16.48}$$

16.5.1 Wall boundary condition

Although the wall boundary condition does not affect the equilibrium of a non-neutral plasma, it plays a critical role for the diocotron modes because these modes are non-axisymmetric. Because the wall is perfectly conducting, the azimuthal electric field must vanish at the wall, i.e., $E_\theta(a) = 0$. This boundary condition is trivially satisfied for the equilibrium because the equilibrium is axisymmetric and $E_\theta = -r^{-1}\partial\phi/\partial\theta$ vanishes everywhere for an axisymmetric field. However, for a non-axisymmetric perturbation, the azimuthal electric field is $\tilde{E}_{\theta,l} = -\mathrm{i}l\tilde{\phi}_l/r$ and so could, in principle, be finite. In order to have the azimuthal electric field vanish at the wall, each finite l mode must therefore satisfy the wall boundary condition $\tilde{\phi}_l(a) = 0$.

16.5.2 The $l = 1$ diocotron mode: a special case

The $l = 1$ mode is a special case because the $1 - l^2$ term in Eq. (16.48) vanishes and the remaining terms have an extremely simple relationship. Specifically, when $l = 1$ Eq. (16.48) reduces to

$$\frac{r}{\Phi}\frac{\mathrm{d}}{\mathrm{d}r}\left(\frac{1}{r}\frac{\mathrm{d}\Phi}{\mathrm{d}r}\right) = \frac{r}{G}\frac{\mathrm{d}}{\mathrm{d}r}\left(\frac{1}{r}\frac{\mathrm{d}G}{\mathrm{d}r}\right), \tag{16.49}$$

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which has the exact solution

$$\Phi = \lambda G, \tag{16.50}$$

where λ is an arbitrary constant. Using Eq. (16.46), this solution can be expressed in terms of the original variables as (White, Malmberg, and Driscoll 1982)

$$\tilde{\phi}_{l=1}(r) = \frac{\omega - \omega_0(r)}{2\omega} Sr, \tag{16.51}$$

where S is a constant. Since Eq. (16.48) is a second-order ordinary differential equation, it must have two independent solutions and Eq. (16.51) must be one of these. The other solution has a singularity at $r = 0$ and is discarded as non-physical (this other solution can be found by assuming it is of the form $\Phi = FG$ and substituting this assumed Φ into Eq. (16.49) to find F). It is now evident that rigid body rotation corresponds to having $G = 0$ for all r since $\omega_0(r)$ is a constant and $\omega = \omega_0(a)$ for a rigid body. If $G = 0$ then Eq. (16.48) shows that the form of Φ is unrestricted and so rigid body rotation can have any perturbation profile. On the other hand, if the rotation has even the slightest amount of shear, then G must be non-zero and the solution is restricted to be of the form specified by Eq. (16.51).

The perfectly conducting wall boundary condition $E_\theta = 0$ implies $\tilde{\phi}_{l=1}(a) = 0$ and so Eq. (16.51) gives the $l = 1$ mode frequency to be

$$\begin{aligned}\omega &= \omega_0(a) \\ &= -\frac{q}{\varepsilon_0 B a^2} \int_0^{r_p} n_0(r') r' \mathrm{d}r' \\ &= -\frac{E_r(a)}{Ba};\end{aligned} \tag{16.52}$$

this has the interesting feature of depending only on $E_r(a)$, the equilibrium radial field at the wall. Since $E_r(a)$ depends only on the total charge per length, the $l = 1$ mode frequency depends only on the total charge per length and so is independent of the radial profile of the charge density. The coefficient S has been chosen so that the *perturbed* radial electric field at the wall is

$$\tilde{E}_r(a) = -\frac{Sa}{2\omega} \left(\frac{\mathrm{d}\omega_0(r)}{\mathrm{d}r} \right)_{r=a}. \tag{16.53}$$

Since $n_0(a) = 0$ by assumption, Eq. (16.42) shows

$$\begin{aligned}\left(\frac{\mathrm{d}\omega_0(r)}{\mathrm{d}r} \right)_{r=a} &= \left(-\frac{2}{r}\omega_0(r) \right)_{r=a} \\ &= -\frac{2\omega}{a}\end{aligned} \tag{16.54}$$

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and so

$$S = \tilde{E}_r(a). \tag{16.55}$$

16.5.3 Energy analysis of diocotron modes

The linearized current density of a diocotron mode is

$$\begin{aligned}\mathbf{J}_1 &= n_1 q \mathbf{u}_0 + n_0 q \mathbf{u}_1 \\ &= n_1 q r \omega_0(r)\hat{\theta} + n_0 q \frac{\mathbf{E}_1 \times \mathbf{B}}{B^2}\end{aligned} \tag{16.56}$$

and the $n_1 q r \omega_0(r)\hat{\theta}$ equilibrium flow term enables the wave energy to become negative. The wave is electrostatic so Eq. (7.56) gives the wave-energy density to be (Briggs, Daugherty, and Levy 1970)

$$\begin{aligned}\bar{w} &= \int_{-\infty}^{t} \mathrm{d}t \left\langle \mathbf{E}_1 \cdot \left(\mathbf{J}_1 + \varepsilon_0 \frac{\partial \mathbf{E}_1}{\partial t} \right) \right\rangle \\ &= \frac{\varepsilon_0}{2} E_1^2 + \int_{-\infty}^{t} \mathrm{d}t \left\langle \mathbf{E}_1 \cdot \mathbf{J}_1 \right\rangle .\end{aligned} \tag{16.57}$$

Unlike the uniform plasma analysis in Section 7.4, here both plasma non-uniformity and wall boundary conditions must be taken into account. The change in system energy $\bar{W}$ due to establishment of the diocotron wave is found by integrating $\bar{w}$ over volume and is

$$\begin{aligned}\bar{W} &= \int \mathrm{d}^3 r \left(\frac{\varepsilon_0}{2} \left(\nabla \cdot \phi_1 \nabla \phi_1 - \phi_1 \nabla^2 \phi_1 \right) + \int_{-\infty}^{t} \mathrm{d}t \, n_1 q r \omega_0(r) E_{1\theta} \right) \\ &= q \int \mathrm{d}^3 r \left(\frac{1}{2} n_1 \phi_1 + r \omega_0(r) \int_{-\infty}^{t} \mathrm{d}t \, n_1 E_{1\theta} \right),\end{aligned} \tag{16.58}$$

where Eq. (16.56) and the perfectly conducting wall boundary condition $\phi_1(a) = 0$ have been used. Using Eq. (7.62) and (16.32), the energy for a given l mode can be written as

$$W_l = \frac{q}{2} \mathrm{Re} \int \mathrm{d}^3 r \left(\frac{1}{2} \tilde{n}_l \tilde{\phi}_l^* \mathrm{e}^{2\omega_i t} + r \omega_0(r) \int_{-\infty}^{t} \mathrm{d}t \tilde{n}_l \tilde{E}_{l\theta}^* \mathrm{e}^{2\omega_i t} \right). \tag{16.59}$$

In order to evaluate this expression, it is necessary to express all fluctuating quantities in terms of the oscillating potential $\tilde{\phi}_l$. Using Poisson's equation in Eq. (16.35) it is seen that

$$\tilde{n}_l = -\frac{l \tilde{\phi}_l}{r B(\omega - l \omega_0(r))} \frac{\mathrm{d}n_0}{\mathrm{d}r} \tag{16.60}$$

and the complex conjugate of the linearized azimuthal electric field is

$$\tilde{E}_{1\theta}^{*} = \left(-\mathrm{i}\frac{l\tilde{\phi}_l}{r}\right)^{*} = \mathrm{i}\frac{l\tilde{\phi}_1^{*}}{r}. \tag{16.61}$$

The wave energy is thus

$$\begin{aligned}
\delta W_l &= -\frac{ql}{2B}\operatorname{Re}\int \mathrm{d}^3 r \left\{ \begin{array}{c} \dfrac{\left|\tilde{\phi}_l\right|^2}{2r(\omega - l\omega_0(r))}\dfrac{\mathrm{d}n_0}{\mathrm{d}r}\mathrm{e}^{2\omega_i t} \\ +r\omega_0(r)\int_{-\infty}^{t}\mathrm{d}t\dfrac{\mathrm{i}l\left|\tilde{\phi}_l\right|^2\mathrm{e}^{2\omega_i t}}{r^2(\omega - l\omega_0(r))}\dfrac{\mathrm{d}n_0}{\mathrm{d}r} \end{array} \right\} \\
&= -\frac{qL}{4B}\operatorname{Re}\int_0^{2\pi}\mathrm{d}\theta\int_0^a \mathrm{d}r \left\{ \left(\begin{array}{c} \dfrac{\mathrm{e}^{2\omega_i t}}{(\omega - l\omega_0(r))} \\ +\omega_0(r)\int_{-\infty}^{t}\dfrac{2\mathrm{i}l\mathrm{e}^{2\omega_i t}\mathrm{d}t}{(\omega - l\omega_0(r))} \end{array} \right)\frac{\mathrm{d}n_0}{\mathrm{d}r} l\left|\tilde{\phi}_l\right|^2 \right\},
\end{aligned} \tag{16.62}$$

where L is the axial length of the plasma. The second term above can be directly evaluated as

$$\begin{aligned}
\operatorname{Re}\int_{-\infty}^{t}\mathrm{d}t\frac{2\mathrm{i}l\left|\tilde{\phi}_l\right|^2\mathrm{e}^{2\omega_i t}}{(\omega - l\omega_0(r))} &= \operatorname{Re}\int_{-\infty}^{t}\mathrm{d}t\frac{2\mathrm{i}l(\omega_r - l\omega_0(r) - \mathrm{i}\omega_i)}{(\omega_r - l\omega_0(r))^2 + \omega_i^2}\left|\tilde{\phi}_l\right|^2\mathrm{e}^{2\omega_i t} \\
&= \int_{-\infty}^{t}\mathrm{d}t\frac{2l\omega_i}{(\omega_r - l\omega_0(r))^2 + \omega_i^2}\left|\tilde{\phi}_l(t=0)\right|^2\mathrm{e}^{2\omega_i t} \\
&= \frac{l}{(\omega_r - l\omega_0(r))^2 + \omega_i^2}\left|\tilde{\phi}_l\right|^2\mathrm{e}^{2\omega_i t} \\
&\to \frac{l}{(\omega_r - l\omega_0(r))^2}\left|\tilde{\phi}_l\right|^2 \text{ as } \omega_i \to 0.
\end{aligned} \tag{16.63}$$

Thus, in the limit $\omega_i \to 0$,

$$\begin{aligned}
\delta W_l &= -\frac{qL}{4B}\int_0^{2\pi}\mathrm{d}\theta\int_0^a \mathrm{d}r\left[\left(\frac{1}{(\omega - l\omega_0(r))} + \frac{l\omega_0(r)}{(\omega - l\omega_0(r))^2}\right)\frac{\mathrm{d}n_0}{\mathrm{d}r}l\left|\tilde{\phi}_l\right|^2\right] \\
&= -\frac{\pi qL}{2B}\int_0^a \mathrm{d}r\frac{\omega}{(\omega - l\omega_0(r))^2}\frac{\mathrm{d}n_0}{\mathrm{d}r}l\left|\tilde{\phi}_l\right|^2.
\end{aligned} \tag{16.64}$$

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For the $l=1$ mode this can be evaluated using Eq. (16.51) to give

$$\begin{aligned}\delta W_{l=1} &= -\frac{\pi q L}{2B}\int_0^a \mathrm{d}r \frac{\omega}{(\omega-\omega_0(r))^2}\frac{\mathrm{d}n_0}{\mathrm{d}r}\left[Sr\left(\frac{\omega-\omega_0(r)}{2\omega}\right)\right]^2 \\ &= -\frac{\pi q L S^2}{8\omega B}\int_0^a \mathrm{d}r \frac{\mathrm{d}n_0}{\mathrm{d}r} r^2 \\ &= -\frac{\pi q L S^2}{8\omega B}\int_0^a \mathrm{d}r \left(-2rn_0(r) + \frac{\mathrm{d}}{\mathrm{d}r}\left[r^2 n_0(r)\right]\right) \\ &= \frac{\pi q L S^2}{4\omega B}\int_0^{r_p} r n_0(r)\mathrm{d}r, \end{aligned} \tag{16.65}$$

where $n_0(r)=0$ for $r_p < r \leq a$ has been used. Finally, using the middle line of Eq. (16.52) this becomes

$$\delta W_{l=1} = -\frac{\varepsilon_0 \pi L a^2 S^2}{4} \tag{16.66}$$

and so the $l=1$ diocotron mode is seen to be a negative energy mode. This result can also be derived by considering the change in system energy that results when the non-neutral plasma is attracted to a fictitious image charge chosen to maintain the wall as an equipotential when the plasma moves off axis (see Assignments 3 and 4).

For $l \geq 2$, the negative energy nature of the diocotron mode can be understood by considering the change in electrostatic energy stored in an isolated parallel plate capacitor when the distance between the parallel plates is varied. The stored energy is $W = CV^2/2 = Q^2/2C$ and Q is constant because the capacitor is isolated. Because increasing the capacitance at constant Q reduces W, bringing the plates together makes available free energy, which can then be used to bring the plates even closer together. The system is therefore unstable with respect to perturbations that tend to increase the capacitance. This is the electrostatic analog of the ideal magnetohydrodynamic flux-conserving situation discussed on p. 315 where the magnetic energy was shown to be $W = \Phi^2/2L$ and Φ was the magnetic flux linked by an isolated inductor L; it was shown that any flux-conserving perturbation that increases inductance makes available free energy for driving an instability.

If all the charge of a non-neutral plasma is assumed to be located at the radius r_p, then the combination of the non-neutral plasma and the wall at radius a effectively constitutes a coaxial capacitor. Gauss' law shows that the radial electric field is $2\pi r\varepsilon_0 E_r = \lambda$, where λ is the charge per length. The voltage difference between the plasma and the wall is therefore $V = -\int_{r_p}^a E_r \mathrm{d}r = \lambda(2\pi\varepsilon_0)^{-1}\ln(a/r_p)$ and the capacitance per length is $C' = 2\pi\varepsilon_0\left[\ln(a/r_p)\right]^{-1}$. Thus, increasing r_p increases the capacitance and decreases the electrostatic energy associated with the fixed amount of charge. However, such an azimuthally symmetric increase of r_p is forbidden because it would rarefy the plasma, an impossibility since the only allowed motion is an $\mathbf{E} \times \mathbf{B}$ drift, which is incompressible for an electrostatic electric field.

The plasma can circumvent this constraint by undergoing an azimuthally periodic incompressible motion that, on average, increases the capacitance. This could be arranged by splitting the system into an even number of equally spaced azimuthally periodic segments and arranging for an incompressible motion where even-numbered segments move towards the wall and odd-numbered segments move away from the wall. The volume between the wall and the plasma would thus be conserved so the motion would be incompressible but, because capacitance scales as the inverse of the distance between a segment and the wall, the increase in capacitance due to the segments moving towards the wall would exceed the decrease in capacitance due to the segments moving away from the wall. Thus, the electrostatic energy of the system would decrease and free energy would be available to drive an instability.

16.5.4 Resistive wall

Let us now assume that the conducting wall has an insulated segment of length L_s with angular extent Δ and that this insulated segment is connected to ground via a small resistor R. The remainder of the wall is directly connected to ground; this is sketched in Fig. 16.2. The wall is thus at or near ground potential and so there are no electric fields exterior to the wall, and in particular there is no radial electric field just outside the wall.

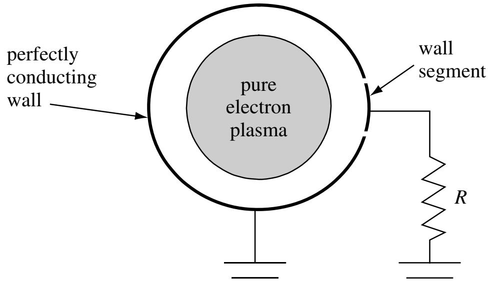

Fig. 16.2 Arrangement for resistive wall instability of a non-neutral plasma. A small resistor R is connected between ground and an isolated wall segment of azimuthal extent Δ and axial extent L_s. The remainder of the perfectly conducting wall is connected directly to ground.

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In order for the radial electric field to vanish outside the wall, the wall must have a surface charge density that establishes a radial electric field equal and opposite to the wall radial electric field S produced by the plasma so

$$-S = \frac{\tilde{\sigma}_l}{\varepsilon_0}. \tag{16.67}$$

The surface charge flows back and forth between the wall segments and ground. The charge on the segment extending from $\theta = -\Delta/2$ to $\Delta/2$ is

$$\begin{aligned} Q_s &= L_s \int_{-\Delta/2}^{\Delta/2} a \mathrm{d}\theta \tilde{\sigma}_l \mathrm{e}^{\mathrm{i}l\theta} \\ &= L_s a \tilde{\sigma}_l \frac{\mathrm{e}^{\mathrm{i}l\Delta/2} - \mathrm{e}^{-\mathrm{i}l\Delta/2}}{\mathrm{i}l} \\ &= -\frac{2}{l} L_s a \varepsilon_0 S \sin\frac{\Delta}{2}. \end{aligned} \tag{16.68}$$

The electric current I flowing through the resistor is the rate of change of this surface charge, i.e., for the $l = 1$ mode this current is

$$I = 2\mathrm{i}\omega L_s a \varepsilon_0 S \sin\frac{\Delta}{2}. \tag{16.69}$$

The rate at which energy is dissipated in the resistor is

$$\left\langle I^2 R \right\rangle = \frac{1}{2}\left(2\omega L_s a \varepsilon_0 S \sin\frac{\Delta}{2}\right)^2 R. \tag{16.70}$$

If the mode frequency is allowed to have a small imaginary part then the rate of change of electrostatic energy will be

$$P = 2\omega_i \delta W. \tag{16.71}$$

Since the sum of the thermal energy in the resistor and the electrostatic energy in the plasma constitutes the total energy in the system, conservation of this sum gives

$$2\omega_i \delta W + \left\langle I^2 R \right\rangle = 0. \tag{16.72}$$

or

$$\omega_i = -\frac{\left\langle I^2 R \right\rangle}{2\delta W}. \tag{16.73}$$

Because the mode has negative energy, ω_i is positive, i.e., the system is unstable.

Substitution into the right-hand side gives

$$\omega_i = \frac{4\omega^2 R L_s^2 \varepsilon_0 \sin^2\dfrac{\Delta}{2}}{\pi L}, \tag{16.74}$$

where L is the axial length of the wall and $L_s \leq L$ is the axial length of the segment. This dependence of the growth rate on resistance has been observed in experiments (White *et al.* 1982). If the resistor R is replaced by a parallel resonant circuit, then the instability will occur at the resonant frequency of this circuit because the effective resistance seen by the non-neutral plasma wall segment will be at a maximum at the resonant frequency. This is essentially the basis for the magnetron tube used in radar transmitters and microwave ovens (see Assignments 5 and 6).

16.5.5 Diocotron modes with $l \geq 2$

The analysis of $l \geq 2$ diocotron modes resembles the Landau analysis of electron plasma waves but is not exactly the same. In general, $l \geq 2$ diocotron modes must be considered using numerical methods because their behavior depends on the equilibrium density profile via coefficients of both the first and last terms in Eq. (16.38). However, some indication of the general behavior can be obtained analytically. An important condition can be obtained (Davidson 2001) by expressing the mode frequency as $\omega = \omega_r + \mathrm{i}\omega_i$ so Eq. (16.38) can be written as

$$\frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}\left(r\frac{\mathrm{d}\tilde{\phi}_l}{\mathrm{d}r}\right) - \frac{l^2}{r^2}\tilde{\phi}_l - \frac{(\omega_r - l\omega_0(r)) - \mathrm{i}\omega_i}{(\omega_r - l\omega_0(r))^2 + \omega_i^2}\frac{l}{rB}\frac{n_0' q}{\varepsilon_0}\tilde{\phi}_l = 0. \tag{16.75}$$

After multiplying through by $r\tilde{\phi}_l^*$, integrating from $r = 0$ to $r = a$, and using the perfectly conducting boundary condition $\tilde{\phi}_l(a) = 0$ when integrating the first term by parts, the following integral relation is found:

$$\int_0^a \mathrm{d}r \left\{ r\left|\frac{\mathrm{d}\tilde{\phi}_l}{\mathrm{d}r}\right|^2 + \frac{l^2}{r}\left|\tilde{\phi}_l\right|^2 + \frac{(\omega_r - l\omega_0(r)) - \mathrm{i}\omega_i}{(\omega_r - l\omega_0(r))^2 + \omega_i^2}\left|\tilde{\phi}_l\right|^2 \frac{l}{rB}\frac{n_0' q}{\varepsilon_0} \right\} = 0. \tag{16.76}$$

The imaginary part of this expression is

$$-\omega_i \frac{lq}{\varepsilon_0 rB}\int_0^{r_p} \mathrm{d}r \left\{ \frac{\left|\tilde{\phi}_l\right|^2}{(\omega_r - l\omega_0(r))^2 + \omega_i^2}\frac{\mathrm{d}n_0}{\mathrm{d}r} \right\} = 0; \tag{16.77}$$

the upper limit has been changed to r_p because $n_0(r)$ and $\mathrm{d}n_0/\mathrm{d}r$ are by assumption zero in the region $r_p < r \leq a$. If $\mathrm{d}n_0/\mathrm{d}r$ has the same sign throughout the radial interval $0 \leq r \leq r_p$, then the integral would have to be non-zero since the integrand always has the same sign, and so ω_i would have to be zero. Thus, a necessary condition for ω_i to be finite is for $\mathrm{d}n_0/\mathrm{d}r$ to change signs in the interval $0 \leq r \leq r_p$. This necessary condition corresponds to $n_0(r)$ having a maximum in the interval $0 \leq r \leq r_p$. This sort of profile is commonly called hollow, because $n_0(r)$ starts at some finite value at $r = 0$, increases to a maximum at some finite $r < r_p$, and then decreases to zero at $r = r_p$.

Further progress can be made by expressing the diocotron equations as a pair of coupled equations for the density and potential perturbations, i.e.,

$$\tilde{n}_l = -\frac{l\tilde{\phi}_l}{rB(\omega - l\omega_0(r))}\frac{\mathrm{d}n_0}{\mathrm{d}r} \tag{16.78}$$

$$\frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}\left(r\frac{\mathrm{d}\tilde{\phi}_l}{\mathrm{d}r}\right) - \frac{l^2}{r^2}\tilde{\phi}_l = -\frac{\tilde{n}_l q}{\varepsilon_0}. \tag{16.79}$$

Rather than substitute for $\tilde{n}_l$ in order to obtain Eq. (16.38), instead Eq. (16.79) is solved for $\tilde{\phi}_l$ using a Green's function method (Schecter *et al.* 2000). This approach has the virtue of imposing the perfectly conducting wall boundary condition on $\tilde{\phi}_l$ at an earlier stage of the analysis before the entire wave equation is developed. The set of solutions to Eq. (16.79) is then effectively restricted to those satisfying the wall boundary condition, and only this restricted set is used when later combining Eqs. (16.79) and (16.78) to form a wave equation.

The Green's function solution to Eq. (16.79) is obtained by first recasting Eq. (16.79) as

$$\frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}\left(r\frac{\mathrm{d}\tilde{\phi}_l}{\mathrm{d}r}\right) - \frac{l^2}{r^2}\tilde{\phi}_l = -\frac{q}{\varepsilon_0}\int_0^a \mathrm{d}s\,\delta(r-s)\tilde{n}_l(s). \tag{16.80}$$

Thus, if $\psi(s, r)$ is the solution of

$$\frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}\left(r\frac{\mathrm{d}}{\mathrm{d}r}\psi(r,s)\right) - \frac{l^2}{r^2}\psi(r,s) = -\delta(r-s), \tag{16.81}$$

where $0 \leq s \leq a$, then

$$\tilde{\phi}_l(r) = \frac{q}{\varepsilon_0}\int_0^a \mathrm{d}s\psi(r,s)\tilde{n}_l(s) \tag{16.82}$$

is the solution of Eq. (16.79). The spatial boundary conditions are accounted for when solving Eq. (16.81) for the Green's function $\psi(r, s)$ and so are independent of the form of $\tilde{n}_l$.

Equation (16.81) is solved by finding separate solutions to its homogeneous counterpart

$$\frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}\left(r\frac{\mathrm{d}\psi}{\mathrm{d}r}\right) - \frac{l^2}{r^2}\psi = 0 \tag{16.83}$$

for the inner interval $0 \leq r < s$ and for the outer interval $s < r \leq a$ and then appropriately matching these two distinct homogeneous solutions at $r = s$ where they meet. The inner solution must satisfy the regularity condition $\psi(0) = 0$ and

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the outer solution must satisfy the wall boundary condition $\psi(a) = 0$. Since the solutions of Eq. (16.83) are $\psi \sim r^{\pm l}$, the inner solution must be

$$\psi = \alpha \left(\frac{r}{a}\right)^l \quad \text{for } 0 \leq r < s \tag{16.84}$$

and the outer solution must be

$$\psi = \beta \left(\left(\frac{r}{a}\right)^l - \left(\frac{r}{a}\right)^{-l}\right) \text{for } s < r \leq a, \tag{16.85}$$

where the coefficients α and β are to be determined.

Integration of Eq. (16.81) across the delta function from $r = s_-$ to $r = s_+$ gives the jump condition

$$\left[\frac{\mathrm{d}}{\mathrm{d}r}\psi(r, s)\right]_{s_-}^{s_+} = -1 \tag{16.86}$$

and integrating a second time shows that ψ must be continuous at $r = s$. These jump and continuity conditions give two coupled equations in α and β,

$$\frac{\beta l}{a}\left(\left(\frac{s}{a}\right)^{l-1} + \left(\frac{s}{a}\right)^{-l-1}\right) - \frac{\alpha l}{a}\left(\frac{s}{a}\right)^{l-1} = -1 \tag{16.87}$$

$$\beta\left(\left(\frac{s}{a}\right)^l - \left(\frac{s}{a}\right)^{-l}\right) - \alpha\left(\frac{s}{a}\right)^l = 0. \tag{16.88}$$

Solving for α and β gives the Green's function,

$$\psi(r, s) = \begin{cases} -\dfrac{a}{2l}\left(\dfrac{s}{a}\right)^{l+1}\left(\left(\dfrac{r}{a}\right)^l - \left(\dfrac{r}{a}\right)^{-l}\right) & \text{for } r > s \\ -\dfrac{a}{2l}\left(\dfrac{r}{a}\right)^l\left(\left(\dfrac{s}{a}\right)^{l+1} - \left(\dfrac{s}{a}\right)^{-l+1}\right) & \text{for } r < s; \end{cases} \tag{16.89}$$

this satisfies Eq. (16.81) and also the boundary conditions at $r = 0$ and $r = a$. Using the Green's function in Eq. (16.82) gives

$$\begin{aligned} \tilde{\phi}_l(r) &= \frac{q}{\varepsilon_0}\left[\int_0^r \mathrm{d}s\psi(r, s)\tilde{n}_l(s) + \int_r^a \mathrm{d}s\psi(r, s)\tilde{n}_l(s)\right] \\ &= -\frac{a}{2l}\frac{q}{\varepsilon_0}\left[\begin{array}{l} \displaystyle\int_0^r \mathrm{d}s\left(\left(\frac{s}{a}\right)^{l+1}\left(\left(\frac{r}{a}\right)^l - \left(\frac{r}{a}\right)^{-l}\right)\right)\tilde{n}_l(s) \\ \displaystyle+\int_r^a \mathrm{d}s\left(\left(\frac{r}{a}\right)^l\left(\left(\frac{s}{a}\right)^{l+1} - \left(\frac{s}{a}\right)^{-l+1}\right)\right)\tilde{n}_l(s) \end{array}\right]. \end{aligned} \tag{16.90}$$

Finally, using Eq. (16.78) to substitute for $\tilde{n}_l(s)$ gives

$$\tilde{\phi}_l(r) = \frac{q}{2\varepsilon_0 B}\left(\begin{array}{c} \displaystyle\int_0^r \mathrm{d}s \frac{s^l}{a^l}\left(\frac{r^l}{a^l} - \frac{a^l}{r^l}\right)\frac{\tilde{\phi}_l(s)}{\omega - l\omega_0(s)}\frac{\mathrm{d}n_0}{\mathrm{d}s} \\ \displaystyle +\int_r^a \mathrm{d}s \frac{r^l}{a^l}\left(\frac{s^l}{a^l} - \frac{a^l}{s^l}\right)\frac{\tilde{\phi}_l(s)}{\omega - l\omega_0(s)}\frac{\mathrm{d}n_0}{\mathrm{d}s} \end{array}\right). \tag{16.91}$$

This integral equation for $\tilde{\phi}_l(r)$ not only prescribes the mode dynamics, but also explicitly incorporates the spatial boundary conditions. The resonant denominators $\omega - l\omega_0(s)$ in the s integrals are reminiscent of the resonant denominators occurring in the velocity-space integrals of the Landau problem. Stability depends on the sign of $\mathrm{d}n_0/\mathrm{d}s$ at the location where $\omega = l\omega_0(s)$ in analogy to the dependence of Landau stability on the sign of $\mathrm{d}f_0/\mathrm{d}v$ where $v = \omega/k$. Because these equations are isomorphic to the 2-D incompressible hydrodynamic equations, it is seen that phenomena similar to Landau damping or instability can occur in 2-D incompressible hydrodynamics.

16.5.6 Phase mixing and relation to Landau damping

The dynamical evolution of an initially off-axis localized infinitesimal bump or patch of increased density provides insight into the vortex dynamics of a two-dimensional inviscid fluid since, as indicated by Eq. (16.22), a region of increased non-neutral plasma density is mathematically equivalent to a region of increased vorticity in a two-dimensional fluid.

If the non-neutral plasma equilibrium angular velocity is sheared, then the off-axis density patch will become sheared because the inner and outer portions of the patch will rotate at different angular velocities. Thus, the angular position of different radial positions of the patch will have trajectories scaling as $\theta(r) = t\omega_0(r)$. The angular separation between two points in the patch starting at respective radii r and $r + \delta r$ will scale as $\delta\theta = t\delta r \mathrm{d}\omega_0/\mathrm{d}r$ and this separation will eventually exceed 2π at sufficiently large t. This gives a sort of spatial phase mixing (Gould 1995) because a localized patch of increased density will eventually be stretched out to become a multi-turn thin spiral of increased density. The number of turns in the spiral increases linearly with time. The patch is thus smeared out over all angles and so is no longer azimuthally localized. Since the patch is stretched azimuthally and yet is incompressible, the thickness of each turn in the spiral must decrease as the number of turns in the spiral increases. Specifically, the length of the spiral increases with t and the radial thickness of each turn decreases as $1/t$ so that the area remains constant. A graphic demonstration of this stretching and thinning has been obtained by Bachman and Gould (1996) using direct numerical integration of the dynamical equations with an initial condition consisting of a prescribed density patch. Since the decay and eventual disappearance of the patch as it deforms into a nearly infinitely long and nearly infinitely thin spiral is a spatial analog to the velocity-space phase mixing underlying Landau damping, it might be expected that the nonlinear mixing of two such patches initiated at two successive times might give rise to a spatial echo effect (Gould 1995). This spatial echo effect involving the nonlinear beating of two spatial spirals has been seen experimentally (Yu and Driscoll 2002). In this experiment an $l = 2$ perturbation is applied and then decays as it is stretched into a nearly infinitely long, nearly infinitely thin spiral. Then an $l = 4$ perturbation is applied, which similarly decays. Finally, after both of the perturbations have decayed, a nonlinear $l = 2$ echo is observed because of nonlinear mixing of the two spirals associated with the respective initial perturbations.

If the patch has a finite amplitude, then its self-electric field will affect the dynamics. This is the nonlinear regime and will cause a sigmoidal curling of the patch since azimuthal electric fields associated with the patch will give radial motions in addition to the azimuthal motions.

16.6 Assignments

1. Relationship between non-neutral and conventional plasma equilibria. Compare the following two plasmas, both of which have cylindrical symmetry, axial uniformity, the same radial electron density profile $n_e(r)$, and are in equilibrium:
   (a) a cylindrical pure electron plasma immersed in a strong magnetic field $\mathbf{B} = B\hat{z}$, where $\omega_{pe}^2 \ll \omega_{ce}^2$;
   (b) an *unmagnetized* quasi-neutral electron–ion plasma, which has infinitely massive ions.

   What is the ion density in the quasi-neutral plasma and what electric field is produced by this ion density? How does the force associated with the electric field produced by the ions in the quasi-neutral plasma compare to the magnetic force associated with electron rotation (finite $u_{e\theta}$) in the pure electron plasma? What is the net equilibrium force on the electrons in the two cases? Assuming that the electrons have zero mass, which plasma has more free energy?
2. Free energy associated with sheared velocity profile in slab approximation. Suppose a non-neutral plasma does not rotate as a rigid body and instead has a sheared angular velocity. An observer rotating with the plasma at some radius r_{obs} would conclude that the plasma has positive angular velocity for $r > r_{obs}$ and a negative angular velocity for $r < r_{obs}$. Since the rotational velocity is due to an $\mathbf{E} \times \mathbf{B}$ drift, this means that the effective radial electric field measured in the observer frame must change sign at $r = r_{obs}$.
   (a) Show that this means that there is an effective charge sheet at $r = r_{obs}$.
   (b) Consider the limiting situation where the above situation is modeled using slab geometry so that $r \to x$ and $\theta \to y$, $\mathbf{B} = B\hat{z}$, and the non-neutral plasma initially centered about $x = 0$ has width w and a uniform density n for $|x| < w/2$. This

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equilibrium is sketched in Fig. 16.3. There is vacuum in the region between the plasma and perfectly conducting walls at $x = \pm a$, where $a > w/2$. What are the electric field and the plasma flow velocity for $x > 0$ and for $x < 0$?

(c) Now suppose that the location of the non-neutral plasma charge sheet is perturbed so that, instead of being centered at $x = 0$, the location of the center becomes a function of y and is given by $x = \Delta(y) = \bar{\Delta}\cos ky$, where $kd \ll 1$ so that the wavelength in the y direction is very long. What is the boundary condition on E_y and hence on ϕ at the walls?

(d) Since there is vacuum between the plasma and the wall and $kd \ll 1$, what is the limiting form of Poisson's equation in the regions between the plasma and the walls?

(e) Assume that w is small, so that the non-neutral plasma can be approximated as being a thin sheet of charge. Being perfectly conducting, the walls must be equipotentials and so, without loss of generality, this potential can be defined to be zero. Show that the potential to the right of the charge sheet must be of the form

$$\phi_r = \alpha(a - x) \tag{16.92}$$

and the potential to the left of the charge sheet must be of the form

$$\phi_l = \beta(a + x), \tag{16.93}$$

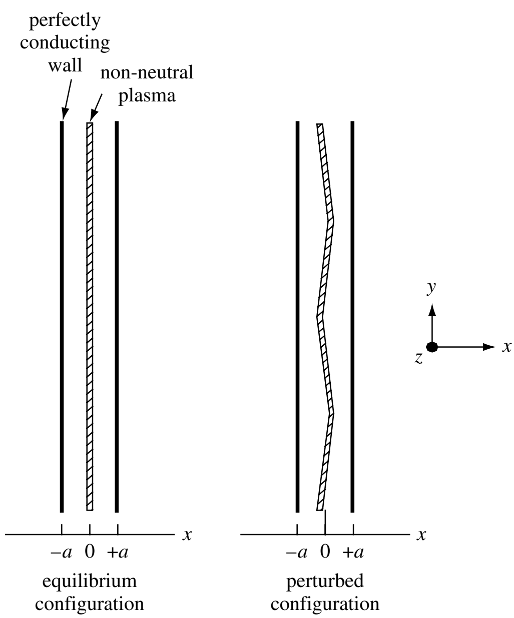

Fig. 16.3 Left: equilibrium configuration for non-neutral plasma of width w, centered between pair of perfectly conducting walls at $x = \pm a$. Right: perturbed configuration where position of x-midpoint of plasma is located at $\Delta = \bar{\Delta}\cos ky$.

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where α and β are coefficients to be determined by considering the jump in electric field at the charge sheet.

(f) Taking into account the jump in the potential at the charge sheet and the continuity in the potential at the charge sheet, solve for α and β. Assume that $w \ll a$ consistent with the assumption that the non-neutral plasma can be considered as a thin charge sheet. Which of E_y^2, E_x^2 is larger?

(g) Suppose that the length of the plasma in the y direction is L, where L is an integral multiple of $2\pi/k$, the wavelength in the y direction. By calculating E_x to the right of the charge sheet and also to the left of the charge sheet, show that the energy stored in the electrostatic electric field is approximately

$$W = \frac{\varepsilon_0}{2} h \int_{-L/2}^{L/2} \mathrm{d}y \int_{-a}^{a} \mathrm{d}x E_x^2,$$

where h is the length in the z direction. How does W change when $\bar{\Delta}$ is increased? What does this suggest about the stability of sheared velocity profiles with respect to perturbations as prescribed in (c)?

3. Image charge in cylindrical geometry. Two line charges λ_1 and λ_2 are aligned along the z axis and located at $\mathbf{r}_1 = x_1\hat{x}$ and $\mathbf{r}_2 = x_2\hat{x}$ as shown in Fig. 16.4. What values of x_2 and λ_2 are required in order for the cylindrical surface $\sqrt{x^2+y^2} = a$ to be an equipotential as would be required in order to have a perfectly conducting wall at $r = a$? The following hints should be useful:

(a) Define a cylindrical coordinate system r, θ, z so that the cylindrical surface is given by $\mathbf{r} = a\hat{r}$. Determine the potential on the cylindrical surface $r = a$.

(b) Show that $|\hat{r} - \hat{x}x_1/a| = |\hat{r}a/x_2 - \hat{x}|$ if $a/x_2 = x_1/a$.

(c) Show that $\lambda_2 = -\lambda_1$ is required in order for the cylindrical surface $r = a$ to be an equipotential.

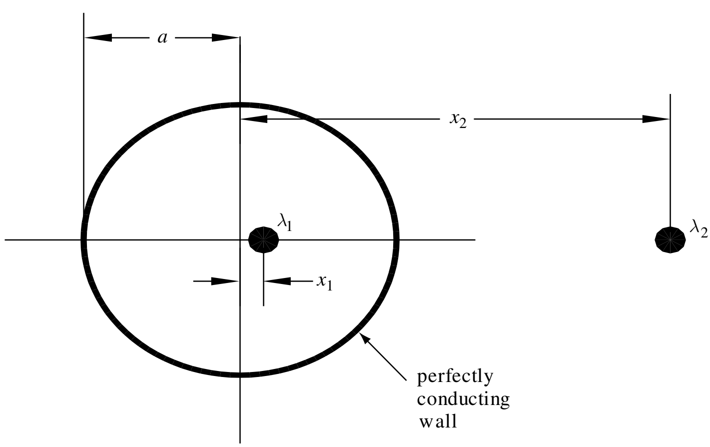

Fig. 16.4 Line charge λ_1 is displaced a distance x_1 from axis of perfectly conducting cylindrical wall of radius a. An image charge λ_2 is located at x_2.

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4. Diocotron mode energy using image charge method. A cylindrical non-neutral plasma is translationally invariant in the z direction and is bounded by a perfectly conducting wall at $r = a$. The plasma has an equilibrium density profile $n = n(r)$ for $0 \leq r < r_p$ and $n = 0$ for $r_p < r \leq a$ so there is a finite-size vacuum region between the edge of the plasma r_p and the wall radius a. Calculate the change in system energy if the plasma axis is shifted off the axis and compare this result to Eq. (16.66). Hints are given below:
   (a) Show that the electric field outside the plasma is the same as the electric field of a line charge located on the plasma axis.
   (b) Suppose that x denotes the distance the plasma is shifted off the wall axis. What is the location and strength of the image charge required for the wall to be an equipotential? Is the force between the plasma and the image charge attractive or repulsive? How does the image charge move as x increases from zero to some finite value?
   (c) Calculate the force on the image charge. What is the force on the plasma? What work is done by the plasma in order to be displaced by a finite amount x? Express this work in terms of the radial electric field perturbation at the wall.
5. Negative energy in terms of two capacitors with fixed charge. Consider the situation where a charge Q is stored on two series-connected parallel plate capacitors that are deformed in such a way as to conserve the total volume of the regions between the plates. This situation is shown in Fig. 16.5(a) top and bottom, and in an electrically equivalent, but geometrically slightly different, way in Fig. 16.5(b) top and bottom.

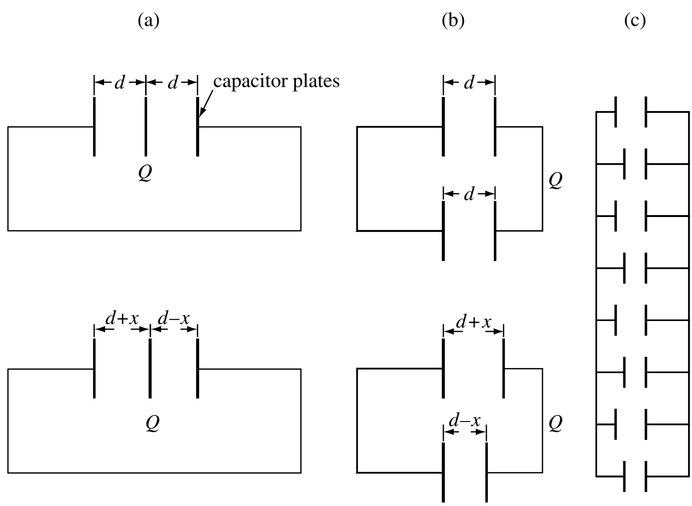

Fig. 16.5 (a) Two capacitors in series with moveable middle plate, (b) geometric rearrangement, (c) straightened-out model of azimuthal periodic incompressible perturbation.

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(a) In Fig. 16.5(a) the center plate is displaced off center by an amount x so that the volume occupied by the two capacitors is conserved. What is the change in the system energy when the center plate is displaced? Assume that the value of each capacitor is $C_{1,2} = \varepsilon_0 A/d_{1,2}$, where $d_{1,2}$ are the distances between the parallel plates and initially (top of figure) $d_1 = d_2 = d$ but later (bottom of figure) $d_1 = d + x$ and $d_2 = d - x$.

(b) Show that the configuration in Fig. 16.5 (b) is electrically equivalent to the configuration in Fig. 16.5(a). Discuss how the system shown in Fig. 16.5(c) would relate to an $l = 4$ diocotron mode (recall that any perturbations to the plasma must be incompressible).

6. Magnetrons as a generalization of non-neutral plasmas undergoing a resistive wall instability. The magnetron vacuum tube used in radar transmitters and in microwave ovens can be considered as a non-neutral plasma undergoing a negative energy diocotron instability. These tubes are simple, rugged, and efficient. The relation between the non-neutral plasma resistive wall instability and the magnetron is shown in the sequence of sketches Figs. 16.6(a)–(d). The magnetron has a cylindrical geometry with an electron emitting filament (cathode) on the z axis, a segmented cylindrical

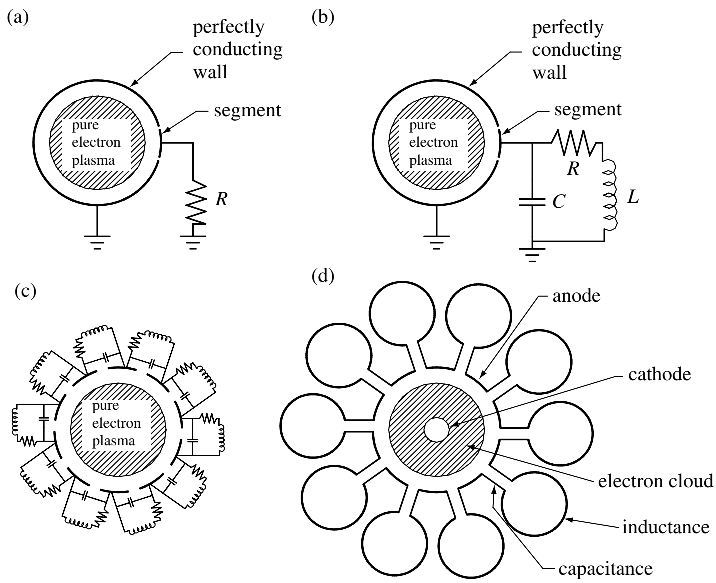

Fig. 16.6 (a) Pure electron plasma with one wall segment connected to ground via a resistor R, (b) same, but now connected to ground via a resonant circuit consisting of a capacitor C in parallel with a coil L and a resistor R, (c) same, but now a set of ten resonant circuits connected across ten gaps, (d) same, but now the coil is a single turn and the capacitance is in the leads connecting to the coil (the resistor is the loading of the coupler to the output circuit, typically a loop inserted into a coil).

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wall (anode) and a z-directed magnetic field produced by permanent magnets. Instead of having a resistor R connected across the gap of a segmented wall as in Fig. 16.6(a), the magnetron has a set of cavities connected across the gap as in Fig. 16.6(d). The cavities function as a resonant circuit and the output circuit loading provides an effective resistance at the cavity resonant frequency. From the point of view of the plasma, the power appears to be dissipated in a resistor across the wall gap. However, since the cavity is coupled to an output waveguide, the power is actually transported away from the magnetron via a waveguide to some external location where the power is used to transmit a radar pulse or cook a meal. If the cavities in Fig. 16.6(d) are phased $0, \pi$ then the ten cavities provide five complete azimuthal wave periods or $l = 5$. If the electrons have near Brillouin flow, and $l = 5$, what axial magnetic field should be used in order to have an output frequency $f = 2450$ MHz, the frequency used in a home microwave oven? Why would the electron density increase to the point that the flow is nearly Brillouin, and why is the electron density not higher than this value? If the voltage drop between anode and cathode is 4 kV and the electron density is uniform, what nominal wall radius (distance between cathode and anode) should be used? Show that a parallel resonant circuit with a small resistance in series with the coil as in Fig. 16.6(b) has an effective resistance that peaks at the resonant frequency; why does one want the effective resistance to peak at the resonant frequency?

7. Spiral due to sheared rotation. Suppose that a non-neutral plasma has a sheared angular velocity so that $\omega_0 = \omega_0(r)$. Suppose that a radial line is painted onto the plasma at $t = 0$ extending from $x = r_p/4$, $y = 0$ to $x = r_p/2$, $y = 0$, where r_p is the plasma radius. Suppose that $\omega_0 = \bar{\omega}_0 r/r_p$. Calculate the trajectory of points on the painted line and plot this for successive times using a computer. If the initial line had finite thickness and hence finite area, what would happen to the thickness of the line with increasing time?